

NAIM NOAMANE

AI Software Engineer — LLM Systems, RAG & Agents, MLOps

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TECHNICAL SKILLS

LLM & Agents: LLM integration (Gemini, Google ADK, MCP), agent orchestration (LangChain, LangGraph, LlamaIndex), prompt engineering, fine-tuning (LoRA, QLoRA, adapters, PEFT, trl, Hugging Face), structured information extraction, PyTorch, TensorFlow
RAG & Retrieval: Qdrant vector store, parent-child chunking, RRF rank-fusion retrieval, BioBERT embeddings, OCR and document processing (Tesseract, OpenCV), Elasticsearch
MLOps / LLMOps & Cloud: MLflow (tracking, versioning, monitoring), Optuna, SHAP, model evaluation, Docker, CI/CD, Azure DevOps, Git / GitHub, Azure, AWS
Backend & APIs: FastAPI, Django, Flask, Streamlit, JavaScript
Data Engineering & Databases: Python (Pandas, NumPy), SQL, Kafka, PostgreSQL, Redis
Machine Learning: XGBoost, Random Forest, scikit-learn, time series (Prophet, AutoARIMA, ETS), causal inference (DoWhy, Granger, DAG)
Languages: Arabic (native), French (fluent), English (fluent)

PROFESSIONAL EXPERIENCE

AI & Data Engineer Intern — Veolia Software Solutions (V2S) | *Marseille, France / Rabat, Morocco* | *Feb 2026 – Jul 2026*

- Designed and industrialised an end-to-end data pipeline on Dataiku DSS, consolidating 12 years of billing and consumption data across many sites (7M rows) into a modelled analytics layer.
- Developed the transformation layer in SQL and custom Python recipes (CTEs, window functions), cutting preparation time per forecasting cycle and enforcing data quality and traceability checks at every stage.
- Integrated climate variables (HDD, CDD, SPI-6) via the Open-Meteo REST API, joining external sources to internal records and improving forecast error.
- Built a forecasting engine (Prophet, AutoARIMA, ETS, Ridge L2) with automated model selection via DTW clustering: 5% MAPE.
- Validated models through causal inference (DoWhy, Granger causality, DAG) and refutability tests (placebo, sensitivity analysis), presented in a Streamlit prototype.

AI & Data Engineer Intern — OCP Group | *El Jadida, Morocco* | *Aug 2025 – Sep 2025*

- Industrialised an agentic recommendation system into production, serving optimised process settings to 10 operators across 2 production units and reducing unplanned shutdowns.
- Designed the LLM reasoning agent with Google ADK and Gemini 2.5 Pro, orchestrating tools exposed through a containerised FastAPI service on Docker.
- Built the ingestion and feature pipeline from Phosphore history (many sensor streams, 3M records): missing-value handling, outlier removal and resampling.
- Trained and tuned supervised models (XGBoost, Random Forest), improving MAPE from 13.5% to 7.5% via Bayesian optimization across 200 Optuna trials.
- Implemented MLflow tracking, model versioning and performance monitoring across many runs, plus SHAP explainability, ensuring traceability and reproducibility for handover to the site team.

Data & Document AI Engineer Intern — ISICOD | *Rabat, Morocco* | *Jul 2024 – Aug 2024*

- Automated extraction of 10 structured fields from scanned and native PDF tender documents using multilingual Tesseract OCR and OpenCV (denoising, adaptive binarization), at 98% field-level accuracy across 30 documents.
- Eliminated a manual review process, cutting tender preparation time, and delivered a retrieval interface used by 4 people.

PROJECTS

AI Gouvernance: LLM API Cost & Accuracy Benchmark | looot.ai

- Designed a benchmark to evaluate API providers' accuracy, reliability, and cost efficiency using verified ground truth.
- Applied statistical testing and cost-per-correct-result analysis to assess provider ranking performance.

LLM Fine-Tuning for Structured Extraction *(LoRA, QLoRA, Adapters, PEFT, Hugging Face)*

- Fine-tuned Qwen2.5 and Llama-3.2 models with LoRA, QLoRA and bottleneck adapters (transformers, peft, trl) on French documents for structured extraction.
- Benchmarked PEFT against full fine-tuning through rank, target-module and data-size ablations, tracked in MLflow; adapters published on Hugging Face Hub.

Medical RAG System *(LangChain, Qdrant, RRF, BioBERT)*

- Built a RAG pipeline over The Gale Encyclopedia of Medicine with parent-child chunking, RRF-based retrieval, BioBERT embeddings (768-d) and a Qdrant vector store, with parent documents retrieved from Google Cloud Storage.
- Generated answers with DeepSeek-R1-Distill-Llama-70B via LangChain and Groq inference, enabling faster, more consistent access to medical knowledge for education, research and support use cases.

EDUCATION

State Engineering Degree in Data Science — School of Information Sciences (ESI) | *Rabat, Morocco* | *Sep 2023 – 2026*
Certifications: Oracle Cloud Infrastructure Foundations Associate (2025) | Agile Scrum Certification (HP) | Databricks Foundations